

AEROSPACE ENGINEERING

Nathan Xia

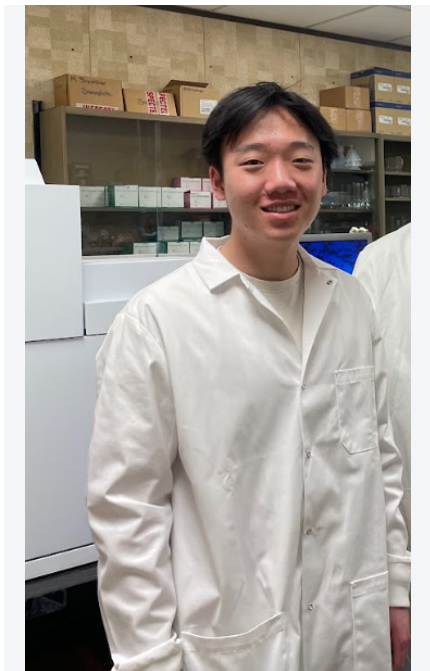
Second-Year Engineering Science Student | University of Toronto

Flow control, aeroacoustics, and experimental & computational fluid dynamics.

ABOUT ME

I’m a second-year Engineering Science student at the University of Toronto, intending to specialize in Aerospace Engineering. My interests centre on flow control and experimental and computational fluid dynamics, with the goal of pursuing graduate research and aerospace R&D.

My work spans aeroacoustic calibration research at UTIAS, aerodynamic design and mechanical leadership with RSX Aerial, and development of an interactive CFD solver. Across these projects, I combine simulation, experimental design, and hands-on manufacturing to investigate flow behaviour and improve engineering designs.



University of Calgary Research  
Research opportunity in the Applied Geochemistry Research Group.

VIEW PROJECTS

DOWNLOAD RESUME

VIEW WEBSITE

TOOLS AND LANGUAGES

|                        |                                                 |
|------------------------|-------------------------------------------------|
| CAD & Manufacturing    | Fusion 360 · SolidWorks                         |
| Simulation             | ANSYS Fluent · Autodesk CFD · OpenVSP / VSPAERO |
| Programming & Controls | Python · MATLAB · C · Java · Arduino IDE        |

SELECTED PROJECTS

|                                              |      |
|----------------------------------------------|------|
| 01 Aeroacoustics Parametric Research Project | 2-4  |
| 02 CanSat Autonomous Glider                  | 5-6  |
| 03 Flow Control Research Project             | 7-8  |
| 04 Interactive 2D CFD Solver                 | 9-10 |

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01 / UTIAS RESEARCH

# Aeroacoustics Parametric Research Project

Undergraduate Research Intern | Summer 2026

## The Problem

RMP calibration changes with probe geometry and test conditions, introducing uncertainty into aeroacoustic measurements.

## Research Scope

- Designed the apparatus and test matrix for a 16-variable parametric study.
- Built a MATLAB acquisition and analysis pipeline for transfer functions and repeatability.
- Developed printed components and a sensor circuit retained by the research group.



### Environmental Sensor Unit

Custom enclosure and display for ambient-condition measurements.

**69% lower TF magnitude standard deviation** with the revised calibration apparatus, relative to the control.

## EXPERIMENTAL CONTEXT

At UTIAS, I independently led a 16-variable study of remote microphone probe (RMP) calibration in Professor Philippe Lavoie’s Flow Control and Experimental Turbulence (FCET) Laboratory. I was one of two first-year students conducting research at UTIAS.

RMPs connect a surface pressure inlet to a remotely located microphone through tubing. Propagation through the tube attenuates the signal and shifts its phase, requiring a transfer-function (TF) calibration. I investigated tube length, diameter, temperature, humidity, and other geometric and environmental factors to understand their influence on calibration consistency.

I owned the workflow from apparatus and test-matrix design through manufacturing, data acquisition, signal processing, statistical analysis, and presentation at the University of Toronto Undergraduate Engineering Research Day (UnERD). My MATLAB pipeline used FFT-based processing to calculate and compare transfer functions across test conditions.

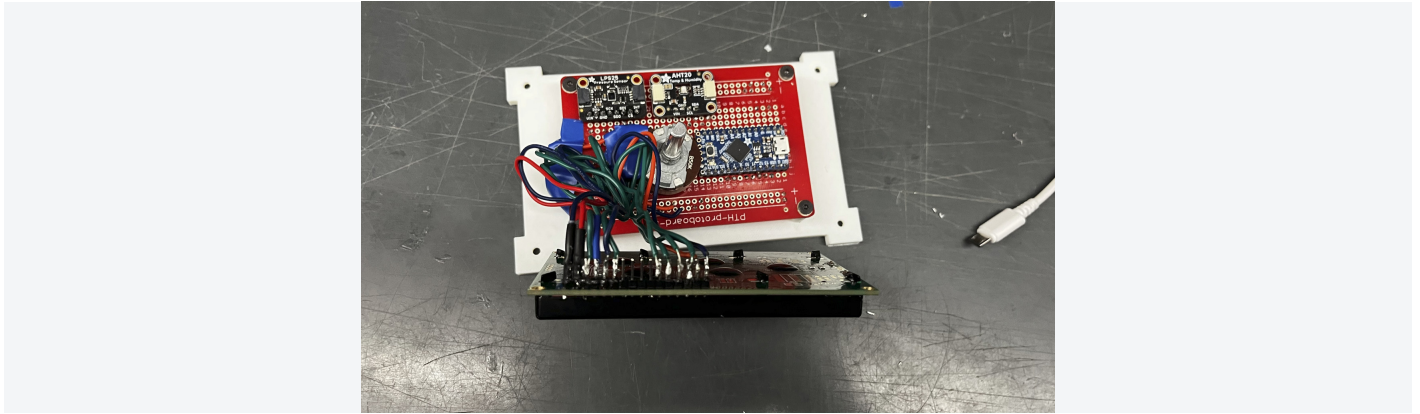
## TOOLS USED

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| Design & Fabrication   | Fusion 360 · SLA and FDM printing / slicing · Soldering and circuit design               |
| Acquisition & Analysis | Arduino IDE · Python · MATLAB · DAQ and data collection · Signal processing and analysis |

## 01 / UTIAS RESEARCH

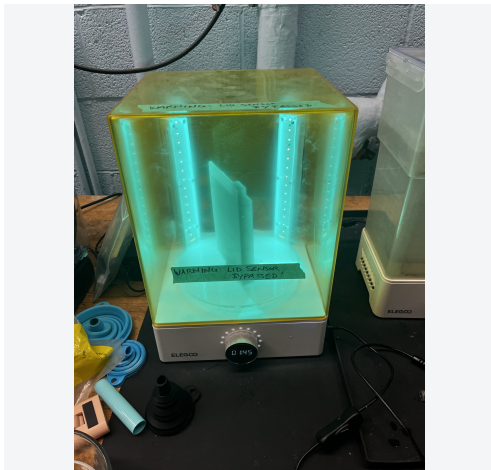
# Instrumentation & Fabrication

I designed SLA-printed components, revised the calibration apparatus, and built an environmental sensor circuit using I2C and SPI. The FCET group retained the apparatus and circuit, alongside data and analysis tools, for subsequent calibration work.



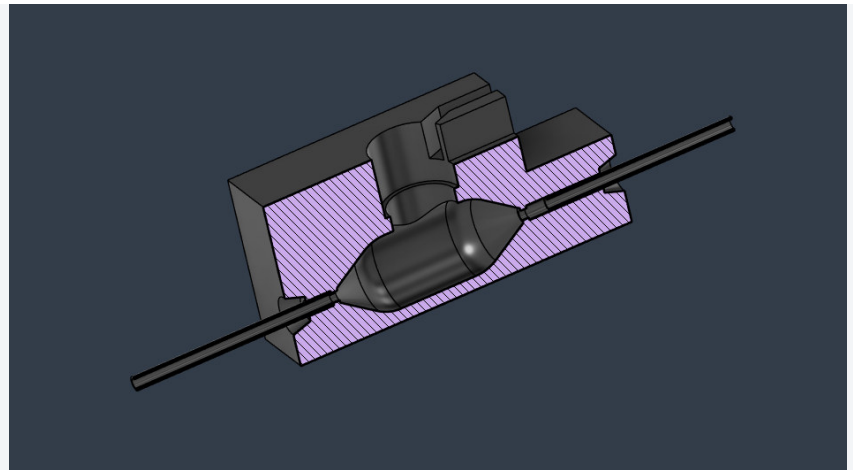
## Sensor Circuit Internals

The sensor unit reads ambient temperature, humidity, and pressure using two sensors communicating over I2C and SPI. An Adafruit ItsyBitsy microcontroller, programmed in Arduino IDE, sends readings to an SPI LCD; a built-in potentiometer adjusts brightness. Sensor readings were manually compared with reference measurements for calibration.



## SLA Printing and Resin Curing

Water-washable resin was selected for fine feature resolution and small layer height. Printed parts were coated with epoxy to seal pores and reduce the risk of acoustic leakage.



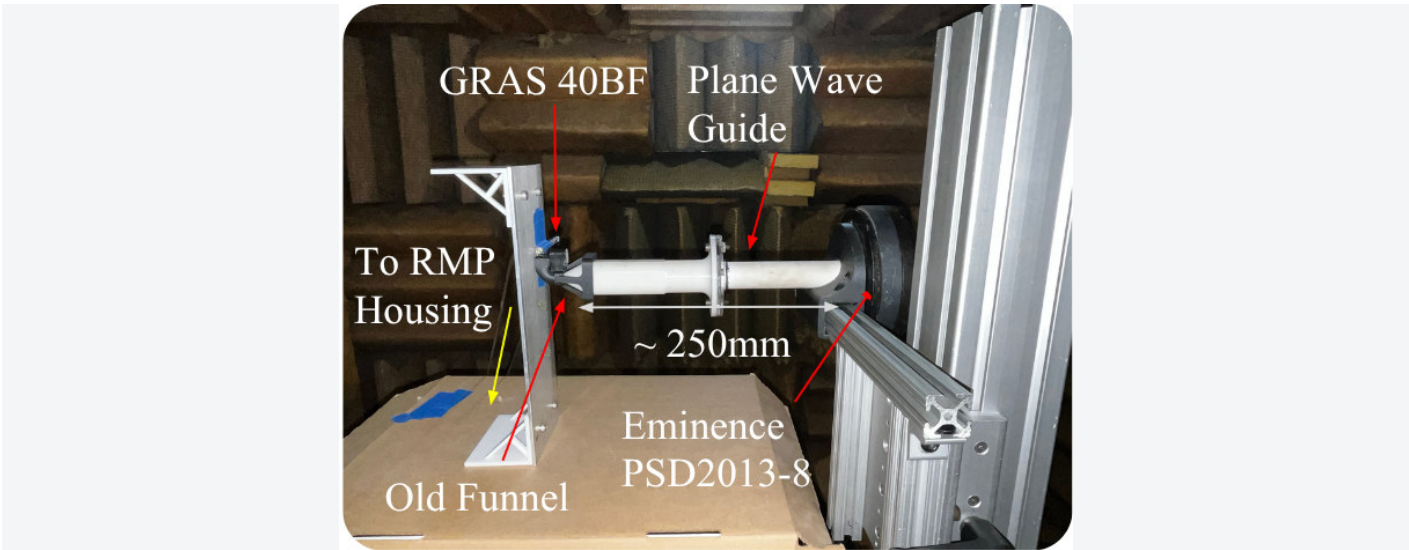
## RMP Housing CAD

The microphone housing was measured from an existing print and redesigned in Fusion 360 to permit geometric modifications.

## RELATING MEASUREMENTS TO PHYSICS

Interpreting the results required relating features in the transfer functions, including attenuation, phase shifts, resonant peaks, and other spectral artifacts, to the underlying acoustic and fluid behaviour within the pneumatic tubing. I compared these observations with analytical models of wave propagation, tube resonance, acoustic impedance, and viscothermal attenuation. This included examining how near-wall boundary layers and other tube-flow effects contributed to frequency-dependent pressure losses.

# Calibration Outcomes



### Calibration Apparatus in the Anechoic Chamber

The labelled setup records signals at 65,536 Hz with a 4,096-sample FFT window. MATLAB handles acquisition and transfer-function analysis.

|                  |                                                                                                                                                                                               |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Repeatability    | 69% lower TF magnitude standard deviation; 33% lower phase standard deviation relative to the control. Values are frequency-averaged across repeated calibrations.                            |
| Research Outputs | Mapped parameter effects to TF behaviour; proposed and experimentally evaluated improvements; delivered a retained sensor circuit, revised apparatus, and analysis tools; presented at UnERD. |

Transfer-function smoothness was assessed using curve tortuosity: the ratio of normalized curve length to the straight-line distance between its endpoints. A value of one represents a straight line. Here, tortuosity describes the plotted transfer function, rather than the physical path of the tube. Repeatability was assessed separately using trial-to-trial standard deviation. See the full report for the methods and comparisons.

### CHALLENGES + SOLUTIONS

**Calibration Repeatability.** Small changes in the relative positions of the microphone, speaker, and probe inlet introduced variability between calibrations. I redesigned the apparatus to hold their positions consistent, reducing transfer-function magnitude standard deviation by 69% relative to the control.

**Interpreting Acoustic Behaviour.** Changes in probe geometry produced complex shifts in attenuation, phase, and resonant features. I compared controlled parameter variations with wave-propagation and acoustic-impedance models to identify physical explanations and inform improved configurations.

**Low-Frequency Coherence.** The original speaker orientation produced poorer coherence at low frequencies. I revised its orientation relative to the reference microphone and probe inlet, improving coherence below 300 Hz.

[Full research paper \(PDF\)](#)

[UnERD poster \(PDF\)](#)

[MATLAB acquisition and analysis repository](#)

02 / RSX AERIAL DESIGN

# CanSat Autonomous Glider

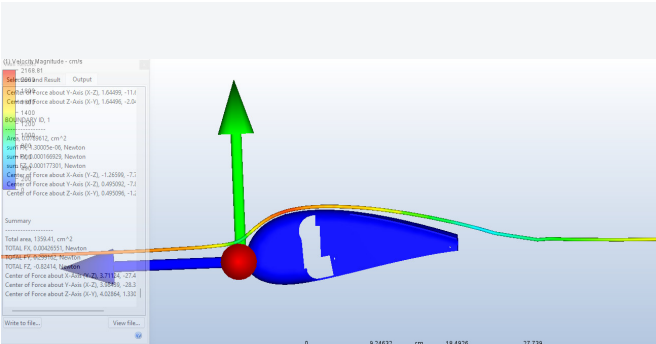
Aerodynamic Design | Mechanical Subsystem | 2025-2026 Competition

## The Problem

Develop a compact, rocket-deployed autonomous glider and integrate its aerodynamic surfaces, structure, and mechanisms.

## My Contributions

- Proposed the selected lifting-body fuselage and screened more than 20 airfoil profiles.
- Refined the fuselage using CFD, vortex generators, and endplates.
- Contributed to composite manufacture, tail components, and mechanical integration.



Glider Body CFD

Vortex generators and endplates address separation and spanwise flow losses in the lifting-body fuselage.

**Approximately 5× simulated fuselage lift** versus the baseline without flow-control devices at 20 m/s.

The fivefold result compares simulated fuselage configurations, not measured lift for the complete aircraft. Experimental confirmation remains a separate step.

## DESIGN AND ANALYSIS WORKFLOW

Within RSX Aerial’s mechanical subsystem, I proposed the lifting-body fuselage concept selected for the final vehicle and led much of its aerodynamic development, from configuration studies through design refinement.

I screened more than 20 airfoil profiles and fuselage geometries using OpenVSP and VSPAERO, then selected promising configurations for Autodesk CFD analysis. Key flow trends and results were cross-checked using ANSYS Fluent. This staged process concentrated detailed analysis on the most promising candidates.

Pressure distributions, streamlines, wake development, and boundary-layer behaviour indicated premature separation and spanwise/tip-flow losses at low Reynolds numbers. Vortex generators and endplates delayed separation in simulation and increased fuselage lift approximately fivefold at 20 m/s.

I also evaluated rocket nosecone geometries in Autodesk CFD and OpenRocket, supported CFD assessment of control mechanisms and aerodynamic interfaces, and assisted with composite wing manufacture. I refined parts for mechanical and electronic integration and manufactured and refined substantial portions of the tail and release mechanisms.

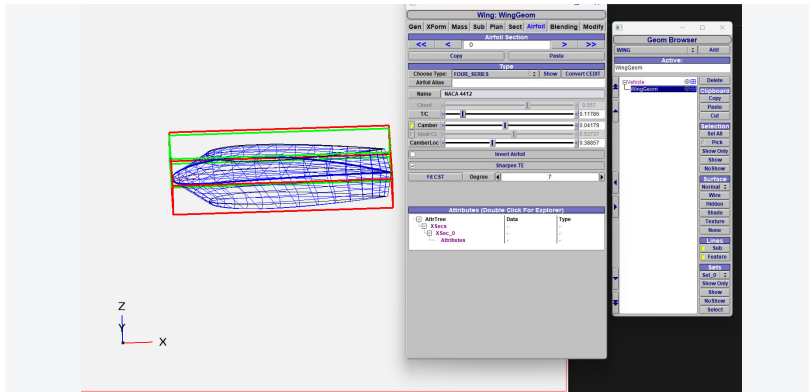
## TOOLS USED

|                     |                                                                           |
|---------------------|---------------------------------------------------------------------------|
| Design & Simulation | Fusion 360 · Autodesk CFD · ANSYS Fluent · OpenVSP / VSPAERO · OpenRocket |
| Build & Integration | Composite manufacturing · Mechanical and electrical integration           |



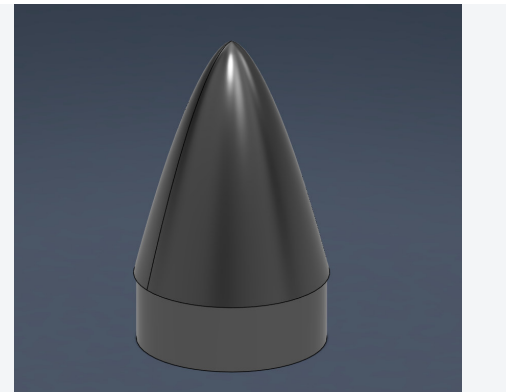
## 02 / RSX AERIAL DESIGN

# Manufacturing & Integration



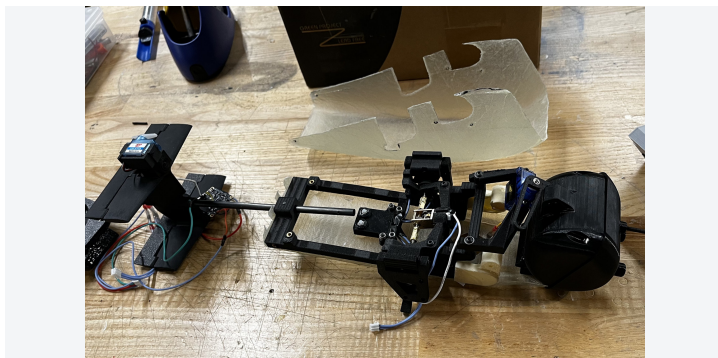
## VSPAERO Screening

Profiles were compared at  $Re \approx 97,000$  using lift coefficient versus angle of attack. Shortlisted profiles included NACA 4421, NACA 4424, and FX 69-PR-281.



## CanSat Nosecone

Power-series nosecone geometries were screened in OpenRocket, then modelled and evaluated with Autodesk CFD to investigate drag.



## TEAM COMPETITION RESULTS

|               |     |
|---------------|-----|
| Overall Score | 70% |
| Canada        | 2nd |
| North America | 8th |

## T-Tail and Final Glider Manufacture

The final vehicle incorporates the fuselage concept I proposed. I contributed to fibreglass/resin composite manufacture, refined tail components for integration, and evaluated tail geometry in CFD. The printed frame used a nylon/carbon-fibre composite.

## CHALLENGES + SOLUTIONS

**Premature Flow Separation.** CFD indicated separation and spanwise losses around the lifting-body fuselage. I incorporated vortex generators and endplates and compared the modified geometry with the baseline, obtaining approximately five times the simulated fuselage lift at the 20 m/s design speed.

**Efficient Geometry Selection.** Evaluating every candidate with detailed CFD would have required substantial iteration time. I used OpenVSP and VSPAERO to screen more than 20 airfoil profiles, then investigated the most promising configurations in Autodesk CFD and cross-checked key flow trends in ANSYS Fluent.

**Manufacturing and Subsystem Integration.** The T-tail and camera integration required reconciling CAD assemblies with manufacturing tolerances and the needs of both the mechanical and electrical subsystems. I refined component clearances and interfaces to support reliable fit and straightforward assembly. For load-bearing printed components, I selected build orientations with the expected loading and weaker interlayer bonds in mind, reducing reliance on interlayer adhesion in critical load paths.

03 / INDEPENDENT EXPERIMENTAL RESEARCH

# Flow Control Research Project

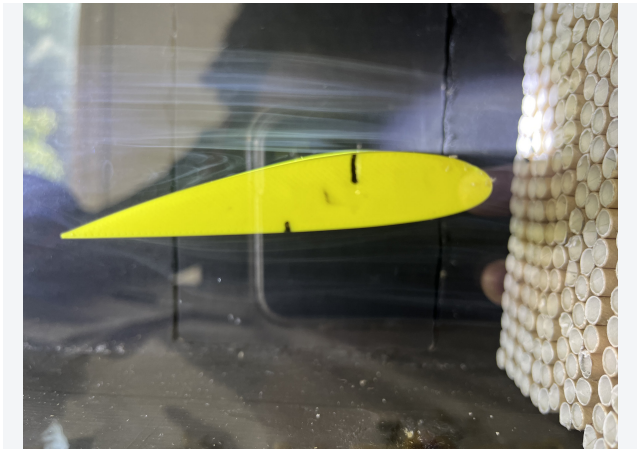
Independent Researcher and Builder | High-School Project | CWSF 2025

## The Problem

Investigate how vortex-generator placement and spacing affect airfoil separation using a wind tunnel built at home.

## What I Designed & Tested

- Built a smoke-visualization wind tunnel and eight 75 mm chord NACA 2412 test airfoils.
- Compared four chordwise positions and four gap spacings, with three trials per configuration.
- Used regression to identify predicted optima and compared them with the untreated airfoil.



### Smoke Visualization

Separation and recirculation around a NACA 2412 airfoil. Incense smoke traces the flow over the wing.

**76% higher predicted separation-criterion angle:**  $6.7^\circ$  for the control versus  $11.8^\circ$  at the fitted optimum.

Visual criterion: separation reaching 30% chord. The fitted optimum is a prediction, not a force-measured stall angle.

## APPARATUS AND EXPERIMENTAL METHOD

This independent project investigated vortex-generator placement and long-gap spacing on a NACA 2412 airfoil. Selected for its relevance to light aircraft such as the Cessna 172, the profile provided my first experience of aerodynamic research and passive flow-control testing.

Without formal laboratory access or dedicated aerospace mentorship, I built a low-speed smoke-visualization tunnel largely from cardboard and hot glue. It operated at approximately 0.84 m/s, with a reported Reynolds number below 2,300. I developed the apparatus, procedure, and analysis using the resources available.

I mathematically generated and 3D printed eight 75 mm chord airfoils. The test matrix compared four locations from 10% to 25% chord and four long-gap spacings from 2 to 5 mm. Separation reaching 30% chord defined the visual criterion as angle of attack increased; each configuration was tested in three trials.

Airfoil profiles were generated in Desmos and converted into STL files in Tinkercad. This was my first experience with CAD.

## TOOLS USED

|                  |                                                                        |
|------------------|------------------------------------------------------------------------|
| Design & Testing | CAD · Desmos · Tinkercad · Wind-tunnel testing and analysis            |
| Methods          | Experimental design, analysis and communication · Passive flow control |

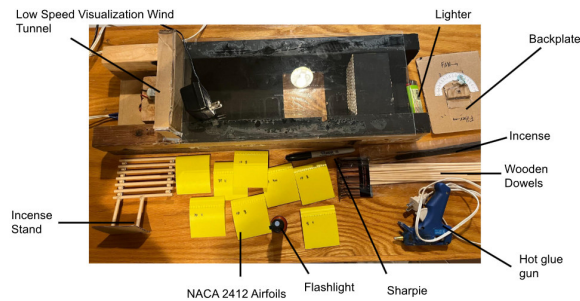
## 03 / INDEPENDENT EXPERIMENTAL RESEARCH

# Experimental Findings



## Custom Suction Fan

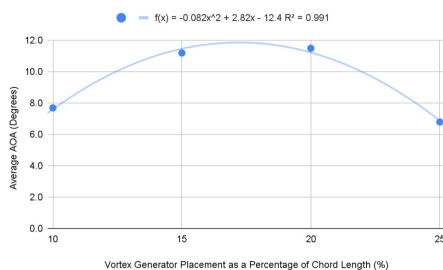
A 3D-modelled replacement inspired by vacuum-cleaner fans supplied the airflow required for visualization.



## Labelled Apparatus

The cardboard-and-hot-glue tunnel, lighting, incense source, angle reference, and eight printed airfoils with varied vortex-generator configurations.

Average Boundary Layer Separation Angle as a Function of Vortex Generator Placement



## Placement-Response Data

The experimental points and parabolic fit identify a predicted optimum in vortex-generator placement.

Placement had a stronger effect than gap spacing. Parabolic fits predicted optima of 17.2% chord and 3.6 mm spacing, with  $R^2$  above 0.936. The fitted placement optimum raised the separation-criterion angle from  $6.7^\circ$  to  $11.8^\circ$  (approximately 76%). Some configurations performed worse than the untreated airfoil, demonstrating the need for deliberate optimization.

These are smoke-based separation measurements and regression predictions, not force-measured stall angles. The low-Reynolds-number experiment does not establish full-scale aircraft performance.

|      |                                                                                                                                        |
|------|----------------------------------------------------------------------------------------------------------------------------------------|
| CYSF | Gold medal; second-best senior project; \$500 award. Selected as one of 15 projects from approximately 600 entries to advance to CWSF. |
|------|----------------------------------------------------------------------------------------------------------------------------------------|

|      |                                                                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| CWSF | Bronze medal among approximately 350 competing projects. Demonstrated passive-flow-control trends and airfoil flow visualization with limited resources. |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------|

## CHALLENGES + SOLUTIONS

**Insufficient Airflow.** The initial fan did not generate enough airflow through the tunnel. I designed and 3D printed a replacement inspired by vacuum-cleaner suction fans, enabling smoke-visualization testing.

**Defining a Repeatable Measurement.** Without lift-measurement equipment, I needed a consistent way to compare airfoil configurations. I defined separation reaching 30% chord as the visual criterion and repeated each configuration across three trials.

**Finding an Effective Configuration.** Adding vortex generators did not consistently improve separation behaviour. I varied placement and gap spacing systematically and used regression to identify predicted optima, rather than assuming that any configuration would be beneficial.

[Full wind-tunnel report \(PDF\)](#)

[CWSF poster \(PDF\)](#)



04 / INDEPENDENT SOFTWARE DEVELOPMENT

# Interactive 2D CFD Solver

Independent Developer | Ongoing Personal Project

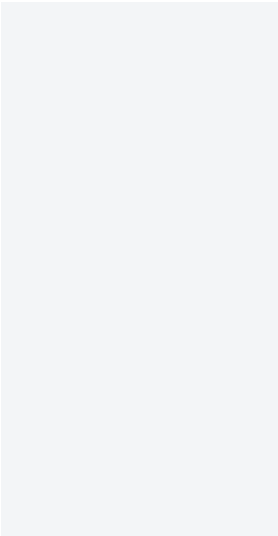
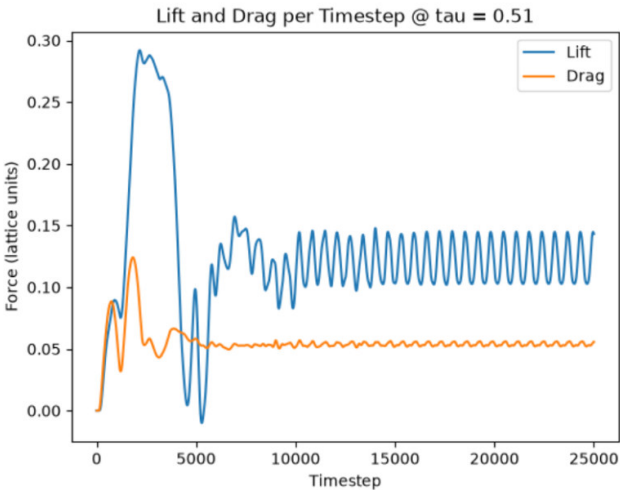
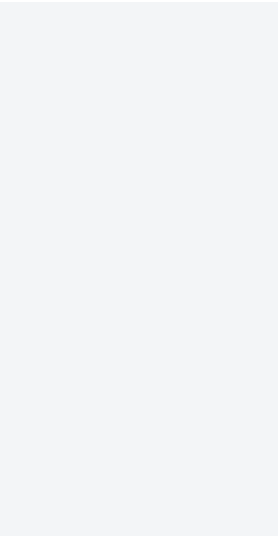
## The Problem

Build an accessible CFD prototype that connects user-drawn geometry with simulated flow and aerodynamic force histories.

## What I Developed

- Implemented a 2D LBM prototype with interactive geometry drawing and autofill.
- Refined boundary conditions and added force extraction.
- Compared NACA 0012 shedding frequency with a published numerical benchmark.

**Approximately 1.8% difference in Strouhal number:** 1.15 from the solver versus 1.1706 in the reference study.



### Simulated Lift and Drag History

The original 25,000-step force history for NACA 0012 at 8° angle of attack and  $Re = 2,000$ . Lift oscillations become periodic after approximately 12,000 steps and provide the shedding-frequency estimate.

### BENCHMARK COMPARISON

|                      |                                                                                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Geometry / Condition | NACA 0012 · Angle of attack 8° · Chord-based Reynolds number 2,000                                                                                               |
| Strouhal Number      | Solver: 1.15   Published reference: 1.1706   Relative difference: approximately 1.8%                                                                             |
| Definition / Scope   | $St = f c / U_\infty$ . The comparison assesses shedding frequency for one operating condition; it does not establish accuracy for all geometries or quantities. |

Reference: D. F. Kurtulus - Vortex flow aerodynamics behind a symmetric airfoil at low angles of attack and Reynolds numbers (Figure 20 and Strouhal-number discussion)

## 04 / INDEPENDENT SOFTWARE DEVELOPMENT

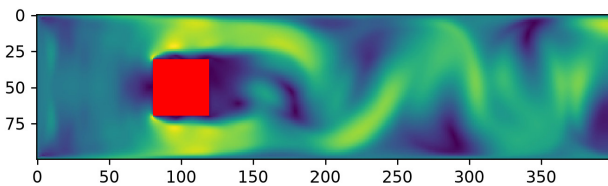
# Solver Development & Evaluation

## CURRENT PROTOTYPE

I am developing an interactive two-dimensional CFD tool using the lattice Boltzmann method (LBM), with the goal of making introductory flow simulation more accessible to beginners, hobbyists, and design teams.

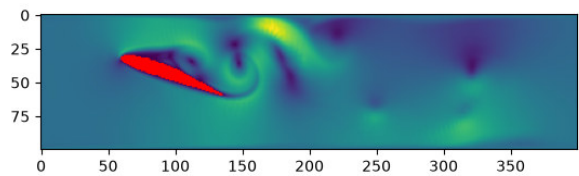
The working prototype allows users to draw a geometry on a grid and simulate flow around it. Implemented features include geometry autofill using breadth-first search, revised inlet and outlet boundary conditions, force extraction, and flow-field visualization. Development has also included collision-operator refinement and a subgrid-model implementation.

The prototype has been exercised with circular and square obstacles and NACA airfoils. A NACA 0012 benchmark compares the frequency of lift oscillations with a published numerical result. The current priority is extending quantitative checks of the solver; a simplified web/cloud interface remains a development goal.



### Square-Obstacle Flow

Velocity is shown from blue (lower) to green/yellow (higher). Alternating von Kármán vortices form behind the square obstacle; the prototype maintains stability around its corners.



### NACA 2412 Velocity Field

Flow visualization at 20° angle of attack shows separation and periodic trailing-edge vortex shedding.

## BENCHMARK INTERPRETATION

For the benchmark shown here,  $St = f c / U_\infty$ , where  $f$  is the shedding frequency,  $c$  is chord length, and  $U_\infty$  is freestream velocity. The solver gives  $St = 1.15$  at  $Re = 2,000$  and an  $8^\circ$  angle of attack. Kurtulus reports  $St = 1.1706$  for this geometry and operating condition using ANSYS Fluent, giving approximately 1.8% relative difference based on the displayed values.

The reference uses ANSYS Fluent. Agreement in shedding frequency supports this specific case; further grid, domain, and force-coefficient checks are needed to establish broader accuracy. Visual flow features are qualitative checks, not substitutes for quantitative evaluation.

## CHALLENGES + SOLUTIONS

**Numerical Stability.** Maintaining stable simulations across different geometries required refinement of the collision operator and boundary conditions. I iterated on these components and tested the prototype with circular, square, and airfoil geometries.

**Quantitative Benchmarking.** Flow visualizations alone could not establish numerical accuracy. I implemented force extraction and used the periodic lift history to estimate shedding frequency, obtaining a Strouhal number approximately 1.8% from the published reference for the NACA 0012 test case.

[View the online project and simulation demonstrations](#)